

DHANA KARTHIKEYA VENTRAPRAGADA

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EDUCATION

University of California, Berkeley

Expected Dec 2027

B.A. in Data Science and Applied Mathematics — GPA: 3.85/4.00

Berkeley, CA

- **Relevant Coursework:** Principles & Techniques of Data Science · Probability for Data Science · Data Structures · *In progress:* Introduction to Machine Learning · Efficient Algorithms & Intractable Problems · Data Engineering

TECHNICAL SKILLS

Languages: Python, SQL, Java

ML: PyTorch, TensorFlow, scikit-learn, XGBoost, LightGBM, SentenceTransformers, Embeddings, Cross-Validation

Data: pandas, NumPy, GeoPandas, Matplotlib, PostgreSQL, Jupyter

Engineering: FastAPI, Gemini API, Pydantic, AWS S3, ffmpeg, pytest, Git

EXPERIENCE

IDX Exchange | Data Science Intern | Remote

Jun 2026 – Aug 2026

- Built a model that predicts what a California home will sell for, with a typical prediction within **8.6%** of the real price, trained on **71K** sales.
- Tested four kinds of models and picked XGBoost, finding that letting it learn more detailed patterns mattered more than training it longer.
- Caught a bug where test data was leaking into training and making early results meaningless, and fixed it so the model was judged fairly.
- Made a simpler model about twice as accurate (error from 29% to 15%) by adding which school district each home is in.

Guang Labs | Engineering Intern | Remote

May 2026 – Aug 2026

- Built a service for an AI video startup that looks at a user's image or video and labels its content type (sports, gaming, talking, music) using Google's Gemini.
- Added a check that catches unsure or broken AI answers and retries with different clips of the video, up to **10** times.
- Got it working end to end after code review by sending real video clips to the AI, and grew its automated tests from 38 to **86**.

PROJECTS

SightLine (Open Projects, ML team) | PyTorch, OpenCV, in progress

- Working on an app that records your day through Meta Ray-Ban smart glasses and automatically edits it into a **60-second** highlight video.
- On the ML team, which splits the footage into clips and uses audio and visual models to pick the best moments of the day.

Movie Genre Classification | Python, scikit-learn, SentenceTransformers

- Built a model that reads a movie's title and plot summary and predicts all its genres, trained on **42K+** films across 20+ genres.
- Improved predictions for rare genres by giving each genre its own cutoff, raising the overall score (Macro F1) from 0.48 to 0.66.
- Found that a simple model beat more complex ones while training about **12×** faster.

Email Spam Classifier | Python, scikit-learn

- Built a spam filter that reads an email's text and flags spam, correctly sorting **92.1%** of emails it had never seen.
- Turned raw emails into 50+ signals, like link counts, dollar signs, and HTML, and kept only the ones that helped.

Housing Price & Assessment Bias | Python, pandas, scikit-learn

- Built a model that estimates home values from **200K+** Chicago-area property records and checked which homes get over- or under-valued.

LEADERSHIP & INVOLVEMENT

UC Berkeley CDSS | Reader/Tutor, CDSS 198 | Berkeley, CA

Aug 2026 – Present

- Run exam reviews for intro CS and data science classes for transfer students, and give resume feedback in a careers seminar.

UC Berkeley EECS | Course Staff (UCS1), CS 61A | Berkeley, CA

Jun 2026 – Aug 2026

- Taught **4** weekly exam-prep sections for Berkeley's intro CS course, wrote exam questions, and built automatic code graders.